The following B.E.S.T Standards will be covered in this unit:
MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions. **MA.7.AR.3.1** Apply previous understanding of percentages and ratios to solve multi-step real-world percent problems. ● *Clarification 1: Instruction includes discounts, markups, simple interest, tax, tips, fees, percent increase, percent decrease, and percent error.*

Unit 12, Lesson 1: Creating Proportions

Benchmark

MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.

Learning Target

☐ I can create proportions to represent a situation.

12.1.1 Warm-Up: Math Flop

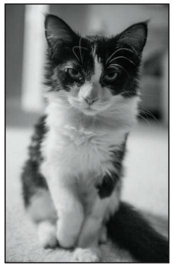
The advertisement encourages making DiGiorno pizza a part of your weekly meals.

1. What do you notice about the advertisement?

12.1.2 Exploration: Proportional Thinking

A photograph is enlarged to make a poster. Measurements are given in centimeters (cm).

16 cm



10 cm

?

25 cm

Abigail used her pencil and measured the photo's width using her finger to mark off 10 cm on the pencil. She then used the pencil to estimate the height using the mark on her pencil.

1. Discuss with your partner why Abigail's strategy could lead to an incorrect answer.

Mariana and Yousef used different strategies to find the height of the poster. Review their solution strategies. Then, work with your partner to answer the questions that follow.

Mariana's Work	Yousef's Work
$\frac{25\ cm}{10\ cm} \div \frac{10}{10} = \frac{2.5}{1} = 2.5$ Scale factor $= 2.5$ $16\ cm \times 2.5 = 40\ cm$ **The height of the poster is 40 cm.**	$\frac{height}{width} \Big\vert \ \frac{16\ cm}{10\ cm} = \frac{?}{25\ cm}$ $\frac{16\ cm}{10\ cm} \times \frac{2.5}{2.5} = \frac{40\ cm}{25\ cm}$ **The height of the poster is 40 cm.**

2. How are Mariana and Yousef's solution strategies similar?

3. How are their solution strategies different?

12.1.3 Bring it Together: Proportional Relationships

In a **proportional relationship**, the values for one quantity are each multiplied by the same number to get the values for the other quantity.

A **proportion** sets two or more equivalent ratios in such relationships as equal to each other and can be used to solve problems.

A ***unit rate*** is a ratio comparing a number of units of one quantity to 1 unit of a second quantity.

1. Use the words and phrases in the bank below to complete the description of Mariana and Yousef's work.

is equal to	proportion	ratio	unit rate

The photograph and the poster are in a **proportional relationship** because the poster is a scaled version of the photograph. Mariana found the scale factor, which is a ________________, and multiplied it by the height of the photo to find the height of the poster. Yousef set up a ________________ stating the ________________ of the height to width of the photograph ________________ the ratio of the height to width of the poster.

12.1.4 Guided Instruction: Proportional Thinking

Proportions can be set up in different ways using the relationships between each quantity.

Four different proportions that can be used to find the height of the poster are shown.

A	**B**
photo to poster $\frac{\text{height}}{\text{width}} \;\Big|\; \frac{16\ cm}{10\ cm} = \frac{?}{25\ cm}$ $\frac{16\ cm}{10\ cm} \times \frac{2.5}{2.5} = \frac{40\ cm}{25\ cm}$	poster to photo $\frac{\text{width}}{\text{height}} \;\Big|\; \frac{25\ cm}{?} = \frac{10\ cm}{16\ cm}$ $\frac{25\ cm}{40\ cm} = \frac{10\ cm}{16\ cm} \times \frac{2.5}{2.5}$
C	**D**
width to height $\frac{\text{photo}}{\text{poster}} \;\Big|\; \frac{10\ cm}{25\ cm} = \frac{16\ cm}{?}$ $\frac{10\ cm}{25\ cm} \times \frac{1.6}{1.6} = \frac{16\ cm}{40\ cm}$	height to width $\frac{\text{photo}}{\text{poster}} \;\Big|\; \frac{16\ cm}{?} = \frac{10\ cm}{25\ cm}$ $\frac{16\ cm}{40\ cm} = \frac{10\ cm}{25\ cm} \times \frac{1.6}{1.6}$

All of the proportions show that the ratios of the dimensions of the photograph and the dimensions of the poster are equivalent.

Complete the statements based on the proportions.

1. The dimensions of the poster are ○ 1.6 ○ 2.5 times larger than the photograph.
2. The height of both the photograph and poster is ○ 1.6 ○ 2.5 times larger than the width.

12.1.5 Guided Practice: Proportional Thinking

The city of Tampa collected sample data on pet ownership to use when determining whether or not it needed to build additional dog parks. A city parks officer randomly sampled 200 households in Tampa and found that 126 of the households had a dog. Using the results of the sample, determine how many households have a dog if there are 153,918 households in Tampa.

1. Complete the statements to describe the relationships that are being compared.

   a. The number of households with a dog is being

      compared to the ___________________________.

   b. The data sample is being compared to the

      ___________________________________ of Tampa.

2. Two proportions that represent this situation are below, but one of them has an error.

Proportion A	Proportion B
$\frac{126}{200} = \frac{153{,}918}{x}$	$\frac{153{,}918}{200} = \frac{x}{126}$

Proportion ○ A ○ B is not set up correctly. The proportion should be ________ = ________.

3. Find the number of households expected to have a dog.

12.1.6 Collaboration: Creating Proportions

1. Javad drove his car to Canada. He was driving on a road and saw the sign shown to the right. According to his speedometer, Javad was driving 75 miles per hour when he was pulled over for speeding. Canadian speed limits are given in kilometers (km) per hour because they use the metric system. There are approximately 1.61 kilometers in 1 mile.

   a. Create two different proportions that could be used to find Javad's speed in kilometers per hour.

Proportion A	Proportion B

   b. How fast was Javad traveling in kilometers per hour?

2. An architect made a scale drawing of a house as shown.

The scale for the drawing is $\frac{1}{4}$ inch $= 6$ feet.

a. Create two different proportions that could be used to find the actual length of the house represented by the drawing.

Proportion A	Proportion B

b. What is the length of the house represented by the drawing?

12.1.7 Wrap-Up: Error Analysis

Jacksonville and Gainesville are approximately 1.4 inches (in.) apart on a scaled map of Florida. The scale used for the map is 2 in. $= 90$ miles (mi).

Three students set up proportions to find the actual distance between Jacksonville and Gainesville, but two of them made errors.

1. For each proportion, check whether it is correct or incorrect. If it is incorrect, fix the error by changing one of the ratios in the proportion.

Lily's Proportion	Eli's Proportion	Mia's Proportion
$\frac{2}{x} = \frac{90}{1.4}$	$\frac{1.4}{2} = \frac{90}{x}$	$\frac{1.4}{x} = \frac{2}{90}$
○ correct ○ incorrect	○ correct ○ incorrect	○ correct ○ incorrect

Unit 12, Lesson 2: Using Proportions to Solve Problems

Benchmark

MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.

Learning Target

- [ ] I can solve problems using proportions.

12.2.1 Warm-Up: Making More Servings

1. Gabriel wants to make 5 servings of oatmeal for breakfast using this recipe. How much water and oats will he need to use? Show your work.

TRADITIONAL STEEL CUT OATS
Hot Water or Milk Preparation

1 Serving

- 1-1/2 cups water or milk
- 1/4 cup Quaker Steel Cut Oats

Directions:

1. Bring water or milk to a boil in a medium saucepan.
2. Stir in oats, reduce heat to low.
3. Simmer uncovered over low heat, stirring occasionally, for 25-30 minutes or until oats are of desired texture.

12.2.2 Guided Instruction: Solving Proportions

Just as there are different ways to set up a proportion to solve a problem, there are different ways to solve the proportion once it's set up.

Jacksonville and Gainesville are approximately 1.4 inches (in.) apart on a scaled map of Florida. The scale used for the map is 2 in. $=$ 90 miles. The proportion below can be used to find the actual distance between Jacksonville and Gainesville.

$$\frac{2}{90} = \frac{1.4}{x}$$

Two methods for solving the proportion are shown.

Method A **Using a Scale Factor**	**Method B** **Solving the Equation**
$\frac{2}{90} = \frac{1.4}{x}$ Determine the factor multiplied by 2 to get 1.4. $1.4 \div 2 = 0.7$ $\frac{2}{90} \times \frac{0.7}{0.7} = \frac{1.4}{63}$	$\frac{2}{90} = \frac{1.4}{x}$ $\overset{1}{\cancel{90}} \cdot \frac{2}{\underset{1}{\cancel{90}}} = \frac{1.4}{x} \cdot 90$ $x \cdot 2 = \frac{126}{\underset{1}{\cancel{x}}} \cdot \overset{1}{\cancel{x}}$ $\frac{\overset{1}{\cancel{2}}x}{\underset{1}{\cancel{2}}} = \frac{126}{2}$ $x = 63$
Both solution methods show the approximate distance between Jacksonville and Gainesville is 63 miles.	

1. In Method A, how does multiplying the ratio $\frac{2}{90}$ by $\frac{0.7}{0.7}$ show that $\frac{2}{90}$ and $\frac{1.4}{63}$ are equivalent?

2. In Method B, why were both ratios multiplied by 90 and then by x?

3. In Method B, why are both sides of the equation multiplied or divided by the same value in each step?

A certain shade of light blue paint is made by mixing $2\frac{1}{2}$ quarts of blue paint with 7 quarts of white paint. To make the same shade of light blue, how much white paint would be needed to mix with 4 quarts of blue paint?

4. Complete the proportion using the information given.

$$\frac{2.5}{\_\_\_} = \frac{7}{\_\_\_}$$

5. With your partner, decide who is Partner A and who is Partner B. Then solve the proportion using the method indicated in the space provided.

Partner A - Method A Using a Scale Factor	**Partner B - Method B** Solving the Equation

6. Check your work with your partner, making any necessary corrections. Once both partners agree, write your partner's method in your workbook. Then complete the statement.

   To make the same shade of light blue, ______ quarts of white paint would be needed.

12.2.3 Collaboration: Examining Student Work

Four students solved different problems involving proportional relationships. Each student's work is shown, including any errors they may have made. Work with your partner or a small group to examine each student's work and complete the table that follows.

Student	Javier
Problem	A mixture of concrete is made up of sand and cement in a ratio of $5:3$. How many cubic feet of sand is needed to make 160 cubic feet of concrete?
Student's Work	$\times 32$ $\frac{5}{3} = \frac{160}{x}$ $\times 32$ $x = 96\ ft^3$

What did Javier do correctly?	**What error(s) did Javier make?**	**Find the correct answer.**

Student	Kayla
Problem	A recipe for making 4 pancakes calls for the ingredients listed. ☐ 6 tablespoons flour ☐ $\frac{1}{4}$ pint milk ☐ 1 pinch salt ☐ 1 egg ☐ $\frac{1}{3}$ pint water You want to make 10 pancakes. a. How much flour do you need? b. How much milk do you need?
Student's Work	a. How much flour do you need? $\cancel{10 : 4 = ? : 6}$ $\cancel{\text{Flour} = \frac{10}{4} = \frac{?}{6}}$ 10 pancakes $= 6 + 6 + 3$ $= 15$ spoonfuls of flour b. How much milk do you need? 10 pancakes $= \frac{1}{4} + \frac{1}{4} + \frac{1}{8} = \frac{3}{16}$ pints

What did Kayla do correctly?	What error(s) did Kayla make?	Find the correct answers.

Student	Isaac
Problem	A drawing of a surfboard in a catalog shows its length as $8\frac{4}{9}$ inches (in.). Find the actual length of the surfboard if $\frac{1}{2}$ in. length on the drawing corresponds to $\frac{3}{8}$ foot of actual length.
Student's Work	$\dfrac{\frac{1}{2}}{8\frac{4}{9}} = \dfrac{\frac{3}{8}}{s}$ $\cancel{8.44} \cdot \dfrac{0.5}{\cancel{8.44}} = \dfrac{0.38}{s} \cdot 8.44$ $s \cdot 0.5 = \dfrac{3.2072}{\cancel{s}} \cdot \cancel{s}$ $\dfrac{\cancel{0.5}s}{\cancel{0.5}} = \dfrac{3.2072}{0.5}$ $s = 6.4$ in.

What did Isaac do correctly?	What error(s) did Isaac make?	Find the correct answers.

Student	Mari
Problem	Juan made 13 out of 20 free throws at basketball practice. If Jada shoots 25 free throws, what is the smallest number of shots she has to make to have a better free-throw percentage than Juan?
Student's Work	$\frac{13}{20} = \frac{f}{25}$ $20 + 5 = 25$ $13 + 5 = 18$ Jada has to make 18 shots.

What did Mari do correctly?	What error(s) did Mari make?	Find the correct answers.

12.2.4 Bring It Together: Proportions

1. Mistakes create opportunities to learn! Based on the mistakes you observed when examining other students' work, use the space provided to describe at least two things you learned and want to keep in mind when using proportions to solve problems.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

12.2.5 Your Turn!

Solve each problem using proportional reasoning. Show your work.

1. Tyler swims at a constant speed of 5 meters (m) every 3 seconds. How long does it take him to swim 114 meters?

2. To make a shade of paint called jasper green, mix 4 quarts (qt) of green paint with $\frac{2}{3}$ cup (c) of black paint. How much green paint should be mixed with 4.5 cups of black paint to make jasper green?

3. It takes an ant farm 3 days to consume $\frac{1}{2}$ of an apple. At that rate, in how many days will the ant farm consume 6 apples?

12.2.6 Wrap-Up: Reflection

1. Draw an **X** on each line to indicate where your current learning is in relation to today's learning target.

   I can solve problems using proportions.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

2. Explain why you placed your **X** where you did.

Unit 12, Lesson 3: Solving Real-World Problems Involving Proportions

Benchmark

MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.

Learning Target

- ☐ I can solve problems using proportions.

12.3.1 Warm-Up: Proportions with Fractions

1. Place a digit in each blue box to make the equality statement true. Use only digits 1 to 9, at most one time each.

$$\frac{\frac{1}{\square}}{\frac{1}{\square}} = \frac{\frac{1}{\square}}{\frac{1}{\square}}$$

12.3.2 Exploration: Which Information Should I Use?

Elena is making large batches of fruit punch and marinated chicken for her upcoming family reunion.

She makes fruit punch by mixing 5 parts cranberry juice with 3 parts apple juice.

1. Complete the statement to explain what this means by writing numbers in each blank.

   Regardless of the unit she uses to measure, Elena will use a ratio of cranberry to apple juice that is ______ to ______. For example, if she is measuring in cups, she would add ______ cups of apple juice for every ______ cups of cranberry juice she adds.

2. Elena plans to make 40 cups of fruit punch. Work with your partner to complete the proportion that could be used to find how much apple juice she needs for the fruit punch, a.

$$\frac{\quad}{\quad} = \frac{a}{40}$$

3. What does the numerator of each ratio represent in the proportion?

4. What does the denominator of each ratio represent in the proportion?

5. How did you use the values 5 and 3 to determine the numerator and denominator of the first ratio?

6. How much apple juice will Elena need to make 40 cups of fruit punch?

The chicken marinade recipe she uses says, "Mix 3 parts oil with 2 parts soy sauce and 1 part orange juice." Elena needs to make 13.5 cups of marinade for the amount of chicken she's marinating.

7. Create a proportion that could be used to find the amount of oil Elena needs for the recipe.

8. Elena is scaling the recipe up to make a larger batch. By what scale factor is Elena enlarging the recipe? Explain or show how you know.

9. How could this scale factor be used to determine how many cups of each ingredient she needs to make the marinade?

10. Complete the summary by writing numbers on each blank based on the situation.

If she's measuring in cups, Elena can make 6 cups of marinade using the original recipe. She is scaling the recipe up by a scale factor of ______ to make 13.5 cups of marinade. To do this, she will mix ______ cups of oil with ______ cups of soy sauce and ______ cups of orange juice.

12.3.3 Bring It Together: Proportions

Recall that ratios can be used to describe part-to-part relationships or part-to-whole relationships.

Situation: Elena's fruit punch recipe uses 5 parts cranberry juice for every 3 parts apple juice.	
Part-to-Part Example	**Part-to-Whole Example**
The ratio of cranberry juice to apple juice is 5:3.	The ratio of cranberry juice to total fruit punch is 5:8.

When setting up proportions to solve problems, it's important to first determine which quantities need to be compared to find the missing value.

1. Complete the table.

Situation: A teacher is planning a field trip to the aquarium. The aquarium requires 2 chaperones for every 15 students. The teacher plans accordingly and orders a total of 85 tickets.	
Set up a proportion that can be used to determine the number of students the teacher is planning to have on the field trip.	
Set up a proportion that can be used to determine the number of chaperones the teacher is planning to have on the field trip.	

2. Solve only one of the proportions above.

3. Explain how the solution to the proportion can be used to determine both the number of student tickets and the number of chaperone tickets the teacher ordered.

12.3.4 Your Turn!

Solve each problem using proportional reasoning. Show your work.

1. A recipe for salad dressing calls for 3 parts oil for every 1 part vinegar. How much oil is needed to make 19 teaspoons (tsp) of salad dressing?

2. A grocer can buy strawberries from a local farmer for $\$1.38$ per pound (lb). His last strawberry order cost $\$241.50$. How many pounds did the grocer order?

3. Andre and Han are moving boxes. Andre can move 4 boxes every half hour. Han can move 5 boxes every half hour. How long will it take Andre and Han to move all 72 boxes?

12.3.5 Wrap-Up: Ticket Out the Door

A recipe for sparkling grape juice calls for $1\frac{1}{2}$ quarts of sparkling water and $\frac{3}{4}$ quart of grape juice.

1. How much sparkling water should be mixed with 9 quarts of grape juice?

2. How much grape juice should be mixed with $\frac{15}{4}$ quarts of sparkling water?

Unit 12, Lesson 4: Discounts and Markups

Benchmark

MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.

Learning Target

☐ I can solve problems involving discounts and markups.

12.4.1 Warm-Up: Civil Engineer

Alishia Ballard works as a civil engineer for San Francisco Public Works. Civil engineers work on the plans to create structures such as bridges, large buildings, etc. They also analyze whether or not structures are safe.

1. How do you think civil engineers use math in their tasks?

Alishia enjoys working at Public Works because it's a way to give back to her community. Every structure plan she works on or structure she inspects is all for the people around her.

2. What is something you have done that gives back to your community, your school, or your peers?

12.4.2 Guided Instruction: Discounts and Markup

Percentages and ratios are often used to express or solve problems involving changes in the price of goods and services.

1. The price of a shirt is $\$18.50$. Natalie has a coupon that reads "discount ½ off."

   a. With the coupon, will Natalie pay more or pay less than $\$18.50$ for the shirt?

   b. Discuss with your partner what "½ off" means. Summarize your discussion.

   A ***discount*** is a reduction of a price for goods or services.

   c. By what percentage will the coupon reduce the price?

d. What is the discount amount, in dollars, taken off the shirt price?

e. What is the discounted price, in dollars, of the shirt after the coupon is applied?

2. Cassidy, a young inventor of Baby Toon, appeared on Shark Tank in 2019. Cassidy shared that it costs her $\$6.60$ to produce one of her baby spoons and that she sells them to customers marked up to $\$15.00$.

a. Does a markup increase or decrease the original price?

b. Discuss with your partner what you think markup means. Summarize the discussion by writing a definition for markup.

c. Complete the statements.

A markup is applied to goods or services so that the seller is able to make a ○ loss. ○ profit. The markup ○ increases ○ decreases the price from the cost to the seller and is usually determined using percentages.

d. If Cassidy placed a 100% markup on her spoons that only cost $\$6.60$ to produce, what would the price of a spoon be for customers?

e. Discuss with your partner how you determined a 100% markup of $\$6.60$. After your discussion, make any revisions to part b, if necessary.

f. Jovi and Malik used two different methods to find the price of the spoon if the markup used is 61%. Part of their work is shown.

Jovi's Work	Malik's Work
$\frac{m}{6.60} = \frac{61}{100}$ $100 \cdot \cancel{6.60}^{1} \cdot \frac{m}{\cancel{6.60}_{1}} = \frac{61}{\cancel{100}_{1}} \cdot \cancel{100}^{1} \cdot 6.60$ $100m = 402.6$ $m = 4.026$	$0.61(6.60) = m$ $4.026 = m$

Complete the statements.

Jovi used a percent ○ equation ○ proportion to find the value of m while Malik used a percent ○ equation. ○ proportion. Both methods resulted in the same value of m, which represents the ○ dollar amount of the markup. ○ price of the spoon after markup.

At a 61% markup, the spoon would sell for ○ $\$4.02$. ○ $\$4.03$. ○ $\$10.62$. ○ $\$10.63$.

g. Jovi and Malik both found the price of the spoon after a 61% markup by calculating the amount of the markup and adding it to the cost of the spoon. What percentage could they have used if they wanted to calculate the price after markup in one step?

12.4.3 Activity: Math Menu

With your partner, choose your meal from the math menu shown. Select one appetizer, one main course, two side dishes, and one dessert. Put a * next to your selections. Then, solve each problem selected.

Appetizers
A1. A book has been discounted by $^{3}/_{10}$. The original price of the book is $\$34.99$. What is the amount being discounted from the book?
A2. Westside Skate Shop is having a storewide $^{1}/_{5}$ off sale. Find the amount saved on a $\$66$ skateboard.
A3. Jennea bought a scratched iPhone because it was discounted $^{9}/_{20}$ off the original price of $\$795$. What was the reduced price, in dollars?
Main Courses
M1. Ramin's Discount Automobiles buys a used car for $\$3,000$. They mark it up 65% before selling it to a customer. What was the sale price of the car?
M2. Elliott sells grab-and-go lunches. He spends $\$2.70$ per lunch on ingredients and then adds a 110% markup to sell. What is the selling price of one sandwich?
M3. It costs JS Surfboards $\$225.00$ in labor to make one Model S surfboard. The company marks up the surfboard 51% to sell. What amount, in dollars, are Model S surfboards sold for?

Side Dishes
S1. A concert ticket is on sale for $\$129$. The sign says 14% off the original price. What was the original price of the ticket?
S2. A scientific calculator is 15% off. The sale price of the calculator is $\$13.99$. What is the original cost of the calculator?
S3. The price of an apple is $\$1.25$. If you get a 10% discount, how much do you have to pay?
S4. Teena bought a bicycle for $\$528$ after getting a discount of 12% taken off its normal price. Find the normal price of the bicycle.
Desserts
D1. A store sells notebooks for $\$1.60$ after a $^{3}/_{10}$ markup. How much is the markup?
D2. Burgermonster sells its burgers for $\$1.84$. To determine the selling price, the restaurant added a markup of $^{23}/_{10}$ to the cost of the ingredients used to make a single burger. What is the cost Burgermonster pays for ingredients?

Appetizer Selection Work and Answer

Main Course Selection Work and Answer

Side Dish Selection Work and Answers

Dessert Selection Work and Answers

12.4.4 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on discounts and markup?

Unit 12, Lesson 5: Determining Tax, Tip, or Fees Given a Percentage

Benchmark

MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.

Learning Targets

- ☐ I can determine the amount of tax, tip, or fee given a percentage.
- ☐ I can determine the percentage of a discount or markup.

12.5.1 Warm-Up: Math Fail

The following sticker was placed on an item for sale.

1. Describe any error(s) on the sticker.

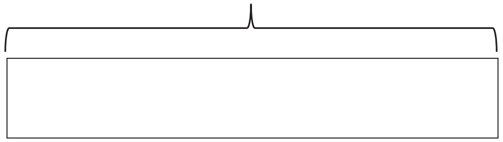
2. Correct the sticker so that it accurately reflects the "$1/2$ Price" advertised in the first line.

12.5.2 Guided Instruction: Finding the Percentage of a Discount or Markup

Recall that discount and markup problems typically include given percentages.

1. The tape diagram shows the original cost of a shirt, $\$30$.

shirt cost ($\$30$)

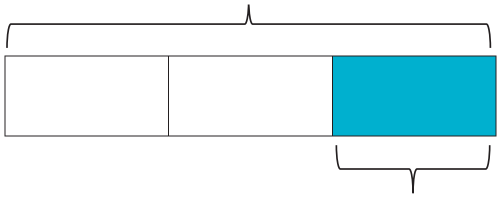


A store manager wants to sell more of this shirt before the end of summer. The manager decides to put the shirt on sale for $1/3$ off the original price.

a. Will customers pay more or less than the original price for the shirt?

The tape diagram is divided into thirds.

shirt cost ($\$30$)



discount

b. What is the price of the shirt after the discount?

c. How much will customers save from the original price?

d. Discuss with your partner how to calculate the percentage discounted if given the original price of the shirt and the sale price of the shirt.

e. If the $\$30$ shirt was discounted to $\$18$, what percentage of the shirt was discounted?

2. Nova buys earrings from a supplier for $\$13.20$. She sells them in her boutique for $\$24.95$. Clare works at the boutique and was curious what percentage markup Nova uses. The table shows 2 different methods Clare used to determine the percentage the earrings are marked up when sold.

Method 1	Method 2
$\$24.95 = (1 + \text{Markup rate}) \times \$13.20$ $\frac{24.95}{13.20} = 1 +$ Markup rate $1.89 = 1 +$ Markup rate $1.89 - 1 =$ Markup rate	Nova paid $\$13.20$ Nova is selling for $\$24.95$ $\$24.95 - \$13.20 = \$11.75$ What $\%$ is 11.75 of 13.2? $\frac{11.75}{13.2} = 0.89$

a. Compare both methods. Discuss the similarities and differences in the two methods with your partner. Then, write them in the table.

Similarities	Differences

b. What is the percentage of the markup?

12.5.3 Try It: Finding the Percentage of a Discount or Markup

1. Johnetta calculated the percentage discounted for a backpack that normally costs $25 but is on sale for $21. Johnetta has made errors in her work. Identify Johnetta's errors and correct her work.

Backpack original cost is $25

Backpack sale cost is $21

$$\frac{\$25}{\$21} = 1.19$$

Discount 1.19%

12.5.4 Guided Instruction: Taxes, Tips, and Fees

1. Baxter has been saving all of his money to purchase new Nike Air Max sneakers. After four months, he finally had $200 saved, so he went to purchase the sneakers. Unfortunately, when Baxter checked out, the total came to $212, which was $12 more than he had saved.

Subtotal	$200.00
Estimated Shipping	$0.00
Estimated Tax	$12.00
TOTAL	$212.00

a. Discuss with your partner why Baxter's total was more than $200.

b. If Baxter had to pay 8% sales tax on the same Nike Air Max sneakers, what would the cost of the taxes be?

c. Explain whether the estimated tax from Baxter's online search was more or less than 8%.

Work with your partner to complete the following.

2. The Cardone Family is planning to have a birthday celebration at one of their favorite restaurants. The restaurants they are considering are in different counties that each have different sales tax rates.

   a. Complete the table to determine how much tax will be at each restaurant given the same pre-tax cost of dinner. The first row of the table has been completed.

Restaurant	Cost of Dinner (Pre-tax)	How Much Sales Tax?
Restaurant A	$\$120$	Sales Tax 7% $(120)(0.07) = 8.40$ $\$8.40$ tax
Restaurant B	$\$120$	Sales Tax 7.5%
Restaurant C	$\$120$	Sales Tax 8%
Restaurant D	$\$120$	Sales Tax 8.5%

   b. The Cardone Family plans to leave a tip for their server. They intend to tip 20% of the cost of their dinner. If the cost of dinner is $\$120$, how much tip will the server receive?

3. Qualyen bought a new computer and printer online totaling $\$850$. He had to pay an additional 3% in shipping fees on top of the $\$850$. With your partner, determine the cost of the shipping fees.

There are many situations where a percentage of an amount of money is added to or subtracted from a total amount. For example:

- If a restaurant bill is $\$34$ and the customer pays $\$40$, they left $\$6$ dollars as a tip for the server. That is 18% of $\$34$, so they left an 18% tip.
- If the price tag on a shirt in Arizona is $\$11.50$, then the sales tax is $(0.075) \cdot \$11.5 = \$0.8625$, which rounds to 86 cents. The customer pays $\$11.50 + \$0.86$, or $\$12.36$ for the shirt.
- A student goes to a hairdresser to get ready for the 8^{th} grade formal dance. The total is $\$125$. The student's parents tip the hairdresser 15%. The total cost of the service is $\$143.75$.
- A gentleman purchases a suit originally priced at $\$397.98$. The suit is on clearance with a 45% off sticker. The discount reduces the suit $\$179.09$ off the original price, resulting in the new suit price being $\$218.89$.

	Goes to	**How It Works**
Sales Tax	The government	Added to the price of the item
Gratuity (Tip)	The person providing the service	Added to the cost of the service
Markup	The seller	Added to the cost of an item so the seller can make a profit
Markdown (Discount)	The customer	Subtracted from the price of an item to encourage the customer to buy it
Commission	The salesperson	Percentage of the value of the sale that is subtracted from the total sale and given to the salesperson

12.5.5 Guided Practice: Calculating Tax, Tip, and Fees

1. Johnny went shopping for his mom at Save More Foods and his subtotal was $\$11.75$. The tax applied at the grocery store is 6%. How much was the tax for his purchase?

2. Rosa and her friends are eating at a restaurant for dinner. Their bill was $\$45.80$. They want to leave a 20% tip. How much tip will the server receive?

3. Rick bought 3 shirts for $\$18$ each, 2 pairs of socks for $\$3.99$ a pair, and a pair of slacks for $\$45.00$. The sales tax rate is 8.5%. How much sales tax did Rick pay?

12.5.6 Your Turn!

1. A city in Ohio has a sales tax rate of 7.25%. Complete the table to determine how much sales tax is for each item.

Item	Price Before Tax ($)	Sales Tax ($)
Pillow	8.69	
Blanket	22.00	
Trash Can	14.50	

2. A store has a sale on pants. The original price of the pants is $\$19.00$. With the discount, the price of one pair of pants before tax is $\$15.20$. What was the percent discount?

3. A music store marks up its instruments. The store bought a guitar for $\$45$ and sold it for $\$58.50$. The store bought a trumpet for $\$80$ and sold it for $\$104$. What is the percent markup the store used to price the guitar and trumpet?

12.5.7 Wrap-Up: Ticket Out the Door

1. Mrs. Jones attended a Southern Living home party. She purchased a vase that cost $\$34.99$ plus $\$3.00$ for shipping. There is a 8.25% sales tax on the cost of the vase and shipping charge.

   What is the amount of tax Mrs. Jones paid on her purchase?

Unit 12, Lesson 6: Solving Multi-Step Problems Involving Discounts, Taxes, Tips, or Fees

Benchmark

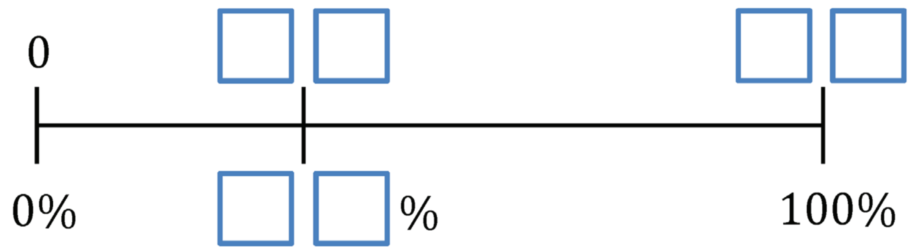
MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.

Learning Target

☐ I can solve multi-step problems that involve discounts, markups, taxes, tips, or fees.

12.6.1 Warm-Up: Percentages on a Linear Model

1. Fill in the blue boxes using the digits $1-9$, one time each at most, to create an accurate number line.



2. There are multiple possible solutions.

   a. Find at least one other solution.

   b. Explain whether using 47% for the percentage makes sense.

12.6.2 Exploration: Finding Sales Tax Rates and Total Amounts

1. Cities often have different sales tax rates based on local government decisions.

   a. Discuss with your partner how you might find the percentage of sales tax if you are given the price of an item and the amount of sales tax paid for the item. Summarize your discussion.

   b. With your partner, complete the two tables to determine the sales tax rates for City A and City B.

City A			
Item	**Price ($)**	**Sales Tax ($)**	**Sales Tax Rate (%)**
Paper Towels	8.99	0.63	
Laundry Soap	11.83	0.83	

City B			
Item	**Price ($)**	**Sales Tax ($)**	**Sales Tax Rate (%)**
Paper Towels	8.99	0.54	
Laundry Soap	11.83	0.71	

   c. Determine the total cost, with tax, of paper towels and laundry soap in City A.

   d. The pre-tax cost of 5 beach towels in City A is $54.95. Determine the total cost, with tax, of 5 beach towels in City A.

   e. If the subtotal for the purchase of a new bathing suit in City B is $47.97, what is the total cost of the bathing suit with tax?

   f. Find the total cost, with tax, on paper towels, laundry soap, and a new bathing suit in City B.

2. Jada and Corrin had lunch at a local restaurant. The final bill for their meal is shown.

They decide to tip their server 18% of the subtotal. Jada says the tip should be $\$7.56$, but Corrin disagrees. Corrin calculated the tip to be $\$8.28$.

Date: Sep. 12th	
Time: 6:55 PM	
Server: # 27	
Bread Stix	9.50
Chicken Parm	15.50
Chef Salad	12.00
Lemon Soda	2.00
Tea	3.00
Subtotal	42.00
Sales Tax	3.99
Total	45.99

a. Explain who calculated the correct tip.

b. What is the total amount, including the tip, that the friends paid at the restaurant?

12.6.3 Collaboration: Taxes, Tips, and Totals

Work through each situation with your partner. Be sure to show your work.

1. Xavier visits an Italian restaurant for dinner.

a. If Xavier tips 20% on the subtotal $\$24.95$, how much tip will he pay?

Date: Sep 19th	
Time: 6:04 PM	
Server: Flo	
Bread Stix	9.50
Ravioli Bites	10.50
Cheesecake	4.95
Subtotal	24.95
Sales Tax	____
Tip	____
Total	____

b. The city has a sales tax rate of 9.5%. How much in tax will the customer pay on the subtotal of $\$24.95$?

c. What is the total Xavier spent on his meal, including tip and tax?

2. A sporting goods store purchased baseball hats from a supplier for $\$24$ each and marked them up to $\$35$.

   a. Calculate the percentage of the markup.

   b. A customer orders 4 baseball hats from the store online. What is the subtotal of the 4 hats?

   c. If the sales tax is 6.25% of the subtotal and there is a $\$7.00$ shipping fee applied after tax, what is the total amount the customer paid?

3. Leilani found a 30% off discount code for an NXT True Hybrid Surfboard. The original cost of the surfboard is $\$700$. If the company does not charge tax but instead charges a $\$150$ shipping fee, how much does the surfboard cost in total?

12.6.4 Your Turn!

1. A car dealership pays a wholesale price of $\$12,000$ to purchase a vehicle. The car dealership wants to make a 32% profit.

   a. How much money is added to the wholesale price to make a 32% profit?

   b. After the markup, what is the total retail price of the vehicle?

   c. The sales tax in the city it is sold is 7.5%. After tax, what will the customer pay for the car?

2. The table shows the pre-tax cost of items at a convenience store where the sales tax rate is 4.6%.

Item	Cost ($)
Milk	3.75
Bag of Ice	2.50
Aspirin – 24 tablets	8.25
Snack Mix	3.75
Candy Bar	1.25
Batteries	6.75

   a. Cal stopped at the convenience store and picked up a bag of ice and batteries. What was his total cost, with tax?

   b. Ms. Isales has $\$10.00$ in her wallet. Does she have enough to buy aspirin and a candy bar? Explain your reasoning using mathematics.

3. A local restaurant automatically adds a 21% gratuity (tip), pre-tax, to each bill for parties over 10. The table shows the bills due for several parties over 10. Complete the table.

Party	Subtotal ($)	Sales Tax ($)	Gratuity ($)	Total Amount of Bill
Flores Party	93.75	5.86		
Atkinson Party	176.99		37.17	
Mikelson Party	135.50			172.43

12.6.5 Wrap-Up: Ticket Out the Door

1. Diesel has been waiting patiently for the Evolve Carbon All Terrain Skateboard to go on sale. For his birthday, he received a 25% off discount. The original price of the skateboard is $\$1,999.99$ and the company has a 5% sales tax, along with a flat fee of $\$35$ for shipping.

   What is the total amount Diesel needs to purchase his dream skateboard?

Unit 12, Lesson 7: Percent Increase and Decrease

Benchmark

MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.

Learning Target

☐ I can find percent increase or percent decrease.

12.7.1 Warm-Up: Interpreting Percentages

1. If exactly 93.6% of people completed a survey, what is the fewest number of people that were surveyed? Consider: How could it be determined if only one person could have completed it? Can partial people complete the survey?

2. Explain your answer using mathematics.

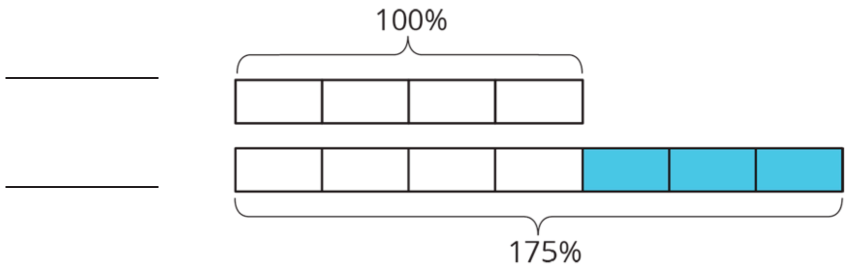
12.7.2 Guided Instruction: Percent of Change

Percentages can be used to make comparisons from one quantity to another.

1. It takes Andre $\frac{3}{4}$ times longer than it took Jada to get to school.

   The diagram models this in terms of percentages.

   a. Label each line of the tape diagram with the name of the student who is represented.



   b. Complete the statements.

   The time it takes Andre to get to school is 75% ○ more ○ less than the time it takes Jada. The diagram shows Andre's travel time is ○ 75% ○ 175% of Jada's travel time.

   c. Let x represent Jada's time. Write an expression to model how long it takes Andre to get to school.

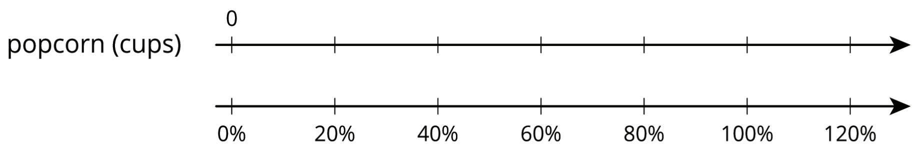
   d. If it takes Jada 8 minutes to get to school, calculate how long it takes Andre to get to school.

The terms "percent increase" and "percent decrease" are used to describe a specific ***percent of change***.

A ***percent of change*** is the difference between a final value and an initial value, expressed as a percentage of the initial value.

A double number line can also be used to model percent increase or percent decrease.

2. A movie theater decreased the size of popcorn bags by 20%. The old bags held 15 cups of popcorn.



   a. On the top number line, plot the 15 cups of popcorn as the initial value of 100%.

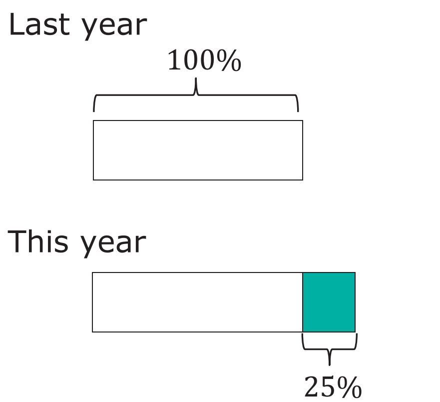
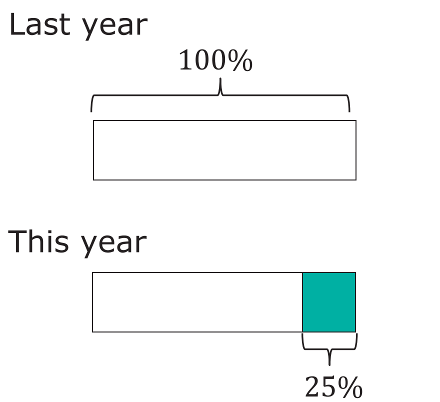
   b. On the second line, plot 20% less than the initial 100%.

c. Use the double number line to find the number of cups in the smaller bag.

d. Discuss with your partner how the double number line is helpful for determining the impact of a 20% decrease in the size of the popcorn bag.

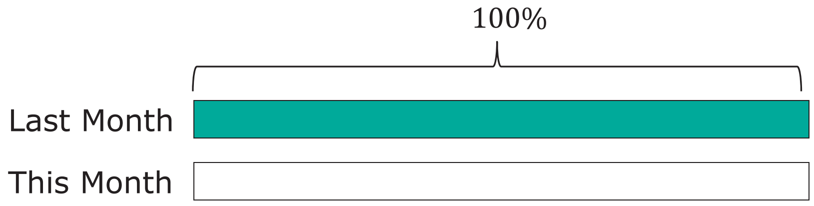
12.7.3 Try It: Percent of Change

1. Explain which diagram, A or B, matches each situation.

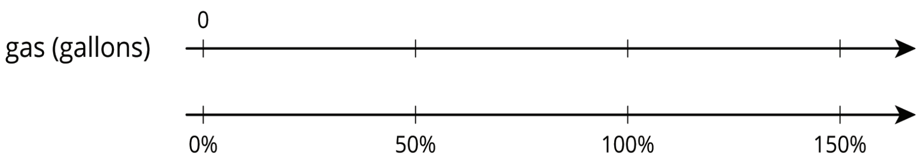
Diagram A	Diagram B
	

Situation	Matching Diagram	Reason
Compared to last year's strawberry harvest, this year's increased 25%.		
This year's blueberry harvest is 75% of last year's.		
Compared to last year, this year's peach harvest decreased by 25%.		
This year's plum harvest is 125% of last year's plum harvest.		

2. The number of mosquitoes decreased by 80% from last month to this month. Create a tape diagram of this month to represent the situation.



3. The gas tank in Robert's truck holds 12 gallons. The gas tank in his mom's truck holds 50% more gas. Using the double number lines, determine how many gallons the gas tank of Robert's mom's truck holds.



12.7.4 Collaboration: Calculating Percent Increase or Decrease

Percent increase and decrease can also be found through calculations.

Complete the following with your partner.

1. A new cereal box contains "20% More" cereal. The original cereal boxes contained 500 grams of cereal.

   a. Explain how to determine how much cereal is in the new box.

   b. How much cereal comes in the new box?

2. The price of a shirt at ClothesMart is $\$18.50$, but there is a store coupon that lowers the price by 20%.

   a. Work with your partner to determine how much the shirt is with the coupon.

The table shows two different ways to complete this problem, along with corresponding answers.

Approach 1	**Approach 2**
The cost of the shirt with the coupon will be 80% of the original cost.	The new cost of the shirt is what's remaining after the 20% has been deducted from the original cost.
New cost $= 0.8(18.50)$	New cost $= 18.50 - 0.2(18.50)$

   b. Explain whether or not these equations are equivalent.

   c. Did your answer to Part A match the solution(s) of the equations in the table?

Percent of Change $= 100 \times \frac{\text{Final Value} - \text{Initial Value}}{\text{Initial Value}}$

3. The original cost of pants at ClothesMart is $\$30.00$, and after a store coupon is applied, the cost is only $\$25.50$.

   a. Determine the percent of change by filling in the blanks of the percent of change ratio for the situation.

$$\frac{\boxed{} - \boxed{}}{Initial} = \frac{\boxed{} - \boxed{}}{\boxed{}} = \frac{\boxed{}}{30} = \boxed{}\ \%$$

   b. Explain whether the percent of change represents a percent increase or decrease based on its value.

   c. Complete the statements.

When calculating the percent of change using the ratio,

a positive value represents a percent
- ○ increase,
- ○ decrease,

and a negative value represents a percent
- ○ increase.
- ○ decrease.

In this context, it makes sense that the percent change is a percent
- ○ increase
- ○ decrease

because the change is the result of a discount.

12.7.5 Your Turn!

1. A school had $1{,}200$ students last year and only $1{,}080$ students this year. What was the percentage decrease in the number of students?

2. A juice box has 20% more juice in its new packaging. The original packaging held 12 fluid ounces. How much juice does the new packaging hold?

12.7.6 Wrap-Up: Error Analysis

1. The number of campers who attend a local summer camp has increased from $1{,}938$ to $2{,}128$. The camp director calculates the percent of increase for an upcoming presentation. Her work is shown.

   $2{,}128 - 1{,}938 = 190$

   $190 \div 2{,}128 = 0.089$

   $0.089 = 8.9\%$

   Explain the camp director's mistake.

Unit 12, Lesson 8: Calculating Percent Error and Simple Interest

Benchmark

MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.

Learning Targets

- ☐ I can find percent error.
- ☐ I can solve problems that involve simple interest.

12.8.1 Warm-Up: Would You Rather?

It's "Multiple Monday" and the more pizzas purchased, the greater the discount applied. If one pizza is ordered, the discount is 10%. If two pizzas are ordered, the discount is 20%, and for three pizzas, the discount goes up to 30%.

1. Would you rather order 3 pizzas on separate checks, or order 3 pizzas on the same check and split the cost?

2. Justify your answer using mathematics.

12.8.2 Exploration: Estimate vs. Actual

1. Instructions to care for a plant say, "Water with $\frac{3}{4}$ cup of water daily." The plant has been getting 25% too much water. How much water has the plant been getting? Include units.

2. The pressure on a bicycle tire is 63 psi. This is 5% higher than what the manual states is the correct pressure. What is the correct pressure? Include units.

3. The crowd at a sporting event is estimated to be $2{,}500$ attendees. The exact attendance is $2{,}486$ people. By what percentage is the estimated attendance off?

4. Discuss with your partner what strategies you used to solve questions $1-3$.

12.8.3 Guided Instruction: Percent Error

Percent error is the difference between the estimated number and the actual number as a percentage of the actual value.

Percent error can be used to describe any situation where there is a correct value and an incorrect value, and to describe the relative difference between them.

1. A teacher estimates there are approximately 600 students at his school, but in actuality there are 625 students enrolled.

   a. What is the difference between the teacher's estimate and the actual number of students enrolled at the school?

   b. Using the definition of percent error, finish filling out the ratio of the situation.

   $$\left|\frac{\text{Expected Value} - \text{Actual Value}}{\text{Actual Value}}\right| = \left|\frac{\quad}{\quad}\right| = 0.\_\_\_\_\_$$

   c. The percent error in the teacher's estimate is _____%.

   d. Discuss with your partner why the percent error is positive. Summarize your discussion.

2. The front of a cereal box lists the net weight of the product is 21 ounces (oz). After further measurements, it is determined there are actually 23 oz of cereal in the box.

   a. What is the difference, in ounces, between the weight listed on the box and the measured weight?

   b. Find the percent error. Show your work.

12.8.4 Guided Practice: Percent Error

1. A store owner estimated she would have 125 customers in her store on its opening day. She underestimated the number of customers by 8%. How many customers visited the store on opening day?

2. To be labeled a jumbo egg, the egg is supposed to weigh 2.5 ounces (oz). Alana buys a carton of jumbo eggs and measures one of the eggs as 2.4 oz. What is the percent error?

3. A student estimates there to be 25.7 milliliters (mL) of a solution in his beaker in science class. He underestimated the amount by 12.5%. What was the actual amount of solution in the beaker? Round to the nearest thousandth.

4. Lia wanted to take advantage of the pizza special on “Multiple Monday.” She estimates it will cost her $\$28$ for two large cheese pizzas after the discount is applied. The actual cost is $\$28.80$. What is the percent error? Round to the nearest whole percent.

12.8.5 Guided Instruction: Simple Interest

1. Kristina received $\$100$ in a birthday card and wants to put the money into her savings account, where she will earn an additional 2% each year it remains in her account. This is known as ***simple interest***.

> ***Simple interest*** is a method of computing interest. Interest is computed from the (original) principal alone, no matter how much money has accrued so far.

   a. How much is 2% of $\$100$?

   b. How much money will be in Kristina's savings account after one year if she makes no additional deposits or withdrawals after putting in her birthday money?

   c. Compare your answer with your partner and discuss how you determined the answer for Part B.

2. Suppose an investment of $\$100$ earns $\$3.50$ interest in one year.

   a. What investment amount will earn $\$61.60$ in two years at the same rate?

   b. Ask two other partner pairs to describe how they found the initial investment. Record their initial investment and write your summary of their process. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Initial Investment	My Summary of Their Reason	Initials

   c. Review the initial investments and reasons provided by other partner pairs. If necessary, fix any mistakes you and your partner made.

Simple Interest Formula

$$I = p \times r \times t$$

where I is the interest earned, p is the principal starting amount of money, r is the interest rate per year, and t is the length of the investment.

3. Tonio took out a six-year car loan for $\$15{,}600$, with a 3% simple interest rate over the life of the loan.

   a. How much interest, in dollars, will Tonio pay annually?

   b. How much will Tonio pay in interest over the course of **six** years?

   c. What is the total amount Tonio will pay back at the end of his six-year loan?

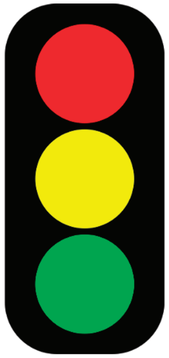
   d. Explain why the total amount Tonio will have paid back by the end of his loan is more than the loan amount he borrowed.

12.8.6 Wrap-Up: Reflection

In this unit, you have learned multiple concepts about percentages and proportional relationships, including those listed below.

- Solving real-world problems using proportions
- Solving discount, markup, tax, tip, and fees problems
- Finding percentage of increase and decrease
- Finding percent error
- Solving problems involving simple interest

1. Use the stoplight reflection below to share how you are feeling about each of the concepts at this point.



I still feel stuck on how to …

I'm starting to get the hang of …

I feel confident and ready to go with …

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